PADHAKU
Exploring Society: India and Beyond · Class 6 Social Science



CHAPTER 1

Locating Places on the Earth

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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Introduction – Why Social Science? · Let's Explore

1 **Observe the picture above. What do you notice?**
Where does the water in the lake come from?
Who made the road and why?
What could be the activities of people living in the small house? What could be their history? Their future?

💡 SAMPLE ANSWER

- I notice a lake, a road, a small house, fields and hills around it.
- The water in the lake comes from rain and from streams flowing down the hills.
- The road was probably made by the government (or the village panchayat) so that people can travel and carry goods to the market.
- The people may farm, keep animals or fish in the lake. Their family may have lived there for many years. In future the road may bring a school, a market or new jobs.

(check the picture in your book)

2 **Write down your answers and discuss them with your classmates.**

💡 SAMPLE ANSWER

Write your answers in your notebook, then compare them with your classmates. Different people may notice different things, and that is fine.

3 **Now, looking at the picture on the facing page, what questions come to your mind? Write them down.**

💡 **SAMPLE ANSWER**

- Who lives in this place and what work do they do?
- How old is this place, and how did it change over time?
- How do people here get water, food and electricity?
- Who looks after the roads and buildings here?

(check the picture in your book)

4 **How do you propose to find answers to the questions related to these two images? How are our questions above relevant to Social Science?**

💡 **SAMPLE ANSWER**

- I can ask elders and people living there, read books and newspapers, look at maps, visit the place, and search on the internet with a teacher's help.

- These questions are about land, water, history, work and government. Social Science studies all these things: how people live, work and are governed, and how places and societies change over time.

⭐ Using a map (Fig. 1.1)

? **You just got off a train at the railway station and want to visit the bank. Which way would you go? Are there other possible ways?**

Can you locate the public garden, the school and the museum?

If you want to go from the bank to the market, which way will you go?

💡 SAMPLE ANSWER

Put your finger on the railway station in Fig. 1.1 and follow the roads to the bank. Usually there is more than one way; the shortest road is the best way.

- Find the public garden, the school and the museum by their pictures and names on the map.
- From the bank, follow the road that leads to the market.

(check the picture in your book)

⭐ Let's Explore · A Map and Its Components

1 **On the map in Fig. 1.1, mark the hospital.**

> 💡 **SAMPLE ANSWER**
> Find the building labelled "Hospital" on the map and draw a circle around it. (check the picture in your book)

2 **What is the meaning of the blue-coloured areas?**

> ✔ **ANSWER**
> Blue colour on a map shows **water**. Here the blue areas show the **lake** (and other water bodies).

3 **Which is farther away from the railway station — the school, the Nagar Panchayat or the public garden?**

> 💡 **SAMPLE ANSWER**
> Measure the distance from the railway station to each place with a ruler; the longest one is the farthest. (check the picture in your book)

4 **As a class activity, form groups of three or four. Each group draws a map of your school, some streets or roads leading to it and a few neighbouring buildings. Compare all the maps and discuss.**

💡 **SAMPLE ANSWER**

- Draw the school in the middle and the roads leading to it.
- Show nearby buildings (shops, houses, temple, bank) with small drawings or symbols.
- Mark north with an arrow.
- When comparing, see which map is clearest and which places each group forgot.

5 **Draw a simple map of a school's playground: a rectangle 40 m long and 30 m wide, on a scale of 1 cm = 10 m.**

Measure the diagonal. How many centimetres do you get? Using the scale, calculate the real length of the diagonal in metres.

✔ **ANSWER**

Length on paper = 40 ÷ 10 = **4 cm**; width = 30 ÷ 10 = **3 cm**.

Draw a rectangle 4 cm × 3 cm.

The diagonal measures **5 cm**.

Real length = 5 × 10 m = **50 m**.

6 **Consider the map of the small city again. Identify the correct and incorrect statements:**

1. The market is north of the hospital.

2. The museum is southeast of the bank.

3. The railway station is northwest of the hospital.

4. The lake is northwest of the apartment blocks.

💡 SAMPLE ANSWER

Use the arrows at the top right of the map: north is up, south is down, east is right, west is left; northwest is up-left and southeast is down-right.

Check each statement this way: start at the second place (hospital, bank, hospital, apartment blocks) and see in which direction the first place lies.

(check the picture in your book)

7 **Taking your school as the starting point, in which cardinal direction is your home approximately located? Discuss with your teacher and parents.**

💡 SAMPLE ANSWER

My home is to the **east** of my school. In the morning, when I walk home from school, I walk towards the rising sun. (Your own answer will depend on where you live.)

8 **Draw a rough map of your locality or village, including your home, school and a few other important landmarks. Show the cardinal directions and use a few symbols from Fig. 1.2.**

💡 SAMPLE ANSWER

- Draw an arrow marked **N** at the top for north (and E, S, W).
- Mark your home, school, a temple/mosque/church, the post office (PO), roads and any river, pond or well.
- Use the Survey of India symbols from Fig. 1.2 (e.g. PO for post office, PS for police station, RS for railway station).

⭐ Let's Explore · a) Understanding coordinates

1 **White opened the game by moving the queen's pawn from d2 to d4. Using the same terms, write down your move if you play black and respond with the same move.**

✔ ANSWER

Black moves the queen's pawn two squares forward: **d7 to d5**.

⭐ Let's Explore · c) Longitudes

1 **Using a globe or atlas, note down approximate values for the latitude and longitude of (1) Mumbai, (2) Kolkata, (3) Singapore, (4) Paris.**

✔ ANSWER

City	Latitude	Longitude
Mumbai	about 19°N	about 73°E
Kolkata	about 22.5°N	about 88°E
Singapore	about 1°N	about 104°E
Paris	about 49°N	about 2°E

⭐ Let's Explore · Understanding Time Zones

1 **Two friends, one in Porbandar (Gujarat) and one in Tinsukia (Assam), are talking on the phone late afternoon. In Assam the sun has set; in Porbandar it is still full daylight. Explain why, and calculate the difference in local time (take the difference in longitude as 30°).**

✔ ANSWER

The Earth spins from west to east, so places in the east see the sun rise and set earlier.

Tinsukia is far to the **east** of Porbandar, so the sun sets there earlier.

The Earth turns 15° in 1 hour.

Difference in local time = 30 ÷ 15 = **2 hours**.
Local time in Tinsukia is about **2 hours ahead** of Porbandar.

2 **Use the example of the two friends in Gujarat and Assam to explain the difference between local time and standard time.**

✔ ANSWER

- **Local time** depends on the position of the sun at a place, so it is different for places at different longitudes. Local time in Tinsukia is about 2 hours ahead of local time in Porbandar.
- **Standard time** is one common time for the whole country. Both friends' clocks show the same Indian Standard Time (IST), even though the sun has set in Assam and not in Gujarat.

? Questions, activities and projects

1 **Using Fig. 5.2 in Chapter 5 and a scale of 2.5 cm = 500 km, calculate the real distance from the estuary of the Narmada River to the estuary of the Ganga river. (Round off your measurement.)**

💡 SAMPLE ANSWER

Scale: 2.5 cm = 500 km, so 1 cm = 500 ÷ 2.5 = **200 km**.

Measured distance on the map ≈ 8 cm (rounded off).

Real distance = 8 × 200 = **about 1,600 km**.

(Measure on the map in your book; your measurement may be a little different.)

2 **Why is it 5:30 pm in India when it is 12 pm or noon in London?**

✔ ANSWER

London is at the Prime Meridian (0°). India's standard time is based on a meridian east of Greenwich (82.5°E).

The Earth turns 15° every hour, so 82.5 ÷ 15 = 5.5 hours.

So Indian Standard Time is **5 hours 30 minutes ahead** of GMT: 12 noon + 5:30 = **5:30 pm**.

3 **Why do we need symbols and colours in the map?**

✔ ANSWER

- There is not enough space on a map to draw every building, road or river, so small symbols are used. Many details fit in a small space.
- Colours show features clearly, e.g. blue for water, green for forests/plains, brown for mountains.

- Standard symbols (like those of the Survey of India) make maps easy for everyone to read.

4 **Find out what you have in the eight directions from your home or school.**

💡 **SAMPLE ANSWER**

Direction	What is there
North	Park
Northeast	Temple
East	Main road and bus stop
Southeast	Market
South	Fields
Southwest	Pond
West	Post office
Northwest	Hospital

(Write what is actually around your home or school.)

5 **What is the difference between local time and standard time? Discuss in groups; each group writes an answer in 100 to 150 words. Compare the answers.**

💡 **SAMPLE ANSWER**

Local time is the time of a place found from the position of the sun. When the sun is highest in the sky, it is 12 noon local time. The Earth spins from west to east and turns 15° every hour, so local time changes by 1 hour for every 15° of longitude. Places to the east are ahead in time. For example, local time in Tinsukia (Assam) is about 2 hours ahead of Porbandar (Gujarat).

If every town used its own local time, trains, offices and TV programmes would be confusing. So a country chooses one meridian and uses its time everywhere. This is called **standard time**. India's standard time (IST) is 5 hours 30 minutes ahead of GMT. Large countries like Russia and the USA have several time zones, each with its own standard time.

6 **Delhi's and Bengaluru's latitudes are 29°N and 13°N; their longitudes are almost the same, 77°E. How much will be the difference in local time between the two cities?**

✔ ANSWER

Local time depends on **longitude**, not latitude. Both cities have the same longitude (77°E), so the difference in local time is **almost zero (no difference)**.

7 **Mark the following statements as true or false; explain your answers with a sentence or two.**

✔ ANSWER

- **All parallels of latitude have the same length. False.** The Equator is the longest; the circles become smaller towards the poles.
- **The length of a meridian of longitude is half of that of the Equator. True.** A meridian is a half-circle from pole to pole, while the Equator is a full circle around the Earth.
- **The South Pole has a latitude of 90°S. True.** The poles are at 90°N and 90°S.
- **In Assam, the local time and the IST are identical. False.** Assam is far east of the meridian used for IST, so its local time is ahead of IST (the sun rises and sets earlier there).
- **Lines separating the time zones are identical with meridians of longitude. False.** They are not straight; they bend to follow international borders.
- **The Equator is also a parallel of latitude. True.** It is latitude 0° and the longest parallel.

8 **Solve the crossword "Locating places on Earth".**

Across: 1. Lets you squeeze a huge area into your map · 4. A convenient sphere · 5. The longest parallel of latitude · 6. The place the

Prime Meridian is attached to · 8. So convenient to find your way · 10. A measure of the distance from the Equator

Down: 2. A measure of the distance from the Prime Meridian · 3. These two together allow us to locate a place · 6. What latitudes and longitudes together create · 7. The time we all follow in India · 9. On top of the world · 11. An abbreviation for a line across which the day and date change

✔ ANSWER

Across: 1. SCALE · 4. GLOBE · 5. EQUATOR · 6. GREENWICH · 8. MAP · 10. LATITUDE

Down: 2. LONGITUDE · 3. COORDINATES · 6. GRID · 7. IST · 9. POLE (North Pole) · 11. IDL (International Date Line)

🎉 **Chapter 1 done! Revise the answers once more tomorrow.**